

AIRLINE FLIGHT DELAY AND PERFORMANCE ANALYSIS

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Tool Stack: Python (Pandas, Matplotlib), MySQL, Power BI, Jupyter Notebook

1. Introduction

The airline industry handles millions of flights every year. Flight delays can negatively impact airline performance, customer satisfaction, and operational efficiency. This project analyzes airline flight data to identify patterns related to delays, airport traffic, airline performance, and weekly flight operations.

The project uses data analytics techniques to transform raw flight data into meaningful insights through Python, SQL, and Power BI.

2. Objectives

The main objectives of this project are:

- To analyze airline flight operations effectively.
- To identify delay patterns and frequencies among different airlines.
- To evaluate traffic volumes across various airports.
- To examine delay trends based on different days of the week.
- To develop an interactive dashboard for effective business decision-making.

3. Dataset Description

The dataset contains airline flight records with operational information.

- **Total Records:** 539,383
- **Total Columns:** 9

Table 1: Dataset Attributes and Descriptions

Column Name	Description
id	Unique flight identifier
Airline	Airline name / carrier code
Flight	Flight number
AirportFrom	Departure airport code
AirportTo	Arrival airport code
DayOfWeek	Day of the week (1-7)
Time	Flight departure time

Column Name	Description
Length	Flight duration in minutes
Delay	Flight delay status (0 = No Delay, 1 = Delayed)

4. Data Preprocessing

Data cleaning was performed rigorously before analysis. The Python Pandas library was used for all preprocessing activities to ensure data integrity.

The following preprocessing tasks were successfully completed:

- Missing value identification and handling.
- Duplicate record detection and removal.
- Data type validation and schema enforcement.
- Final dataset structure verification.

5. Exploratory Data Analysis (EDA)

Several analytical questions were explored to understand the dataset's underlying structure:

- **Total Flights & Airlines:** The dataset contains 539,383 flight records across 18 distinct airlines.
- **Delay Analysis:** Flight delays were analyzed to determine the overall percentage of delayed flights and identify airlines with the highest delay rates.
- **Weekly Delay Analysis:** Flight delays were evaluated across different days of the week to identify operational performance patterns.
- **Airport Traffic Analysis:** Departure and arrival airport traffic were analyzed to extract and identify the top 10 busiest airports in the dataset.

6. SQL Analysis

Structured Query Language (SQL) queries were executed to retrieve crucial business insights from the relational database management system.

Query 1 – Total Flights

SQL

```
SELECT COUNT(*) AS Total_Flights
FROM flights;
```

Query 2 – Total Airlines

SQL

```
SELECT COUNT(DISTINCT Airline) AS Total_Airlines
FROM flights;
```

Query 3 – Delay Rate by Airline

SQL

```
SELECT Airline, AVG(Delay) * 100 AS Delay_Rate
```

FROM flights

GROUP BY Airline

ORDER BY Delay_Rate DESC;

Query 4 – Delay Rate by Day

SQL

SELECT DayOfWeek, AVG(Delay) * 100 AS Delay_Rate

FROM flights

GROUP BY DayOfWeek;

Query 5 – Top 10 Departure Airports

SQL

SELECT AirportFrom, COUNT(*) AS Total_Flights

FROM flights

GROUP BY AirportFrom

ORDER BY Total_Flights DESC

LIMIT 10;

Query 6 – Top 10 Arrival Airports

SQL

SELECT AirportTo, COUNT(*) AS Total_Flights

FROM flights

GROUP BY AirportTo

ORDER BY Total_Flights DESC

LIMIT 10;

7. Dashboard Development

Power BI was utilized to create an interactive, enterprise-grade business intelligence dashboard. The dashboard features a clean user interface with a "Deep Teal" and "Amber" color profile tailored for the aviation industry.

The dashboard structure includes:

- **KPI Cards:** Displaying Total Flights, Total Airlines, Delay%, and Delayed Flights count using advanced Card (New) visuals.
- **Interactive Filters (Slicers):** Select Airline, Select Day of Week, Departure Airport, and Arrival Airport dropdown menus.
- **Visualizations:**
 - *Weekly Delay Analysis* (Horizontal Bar Chart)
 - *Flight Delay Distribution* (Pie Chart)
 - *Flight Volume by Airline* (Column Chart)
 - *Top 10 Departure & Arrival Airports* (Column Charts)

8. Findings

The analysis revealed several important insights regarding flight operations:

- The dataset contains a substantial volume of records, exceeding half a million flights.
- Certain airlines experience disproportionately higher delay rates compared to others.
- Flight delays fluctuate noticeably across different days of the week, indicating peak operational bottlenecks.
- Specific hub airports handle significantly higher flight volumes, requiring optimal resource allocation.
- The delay distribution indicates that a substantial portion (44.54%) of flights experience operational delays.

9. Recommendations

Based on the data-driven insights, the following strategic recommendations are proposed:

- **Optimize Scheduling:** Improve scheduling efficiency and buffer times for high-delay airlines.
- **Operational Monitoring:** Increase monitoring and ground support during peak traffic periods identified in the weekly analysis.
- **Resource Allocation:** Allocate additional resources and traffic management tools to the top busy airports (e.g., ATL, ORD, DFW).
- **Predictive Frameworks:** Implement predictive analytics models to foresee and mitigate potential delay factors.
- **Continuous Auditing:** Continuously monitor real-time airline performance metrics using the developed Power BI dashboard reports.

10. Conclusion

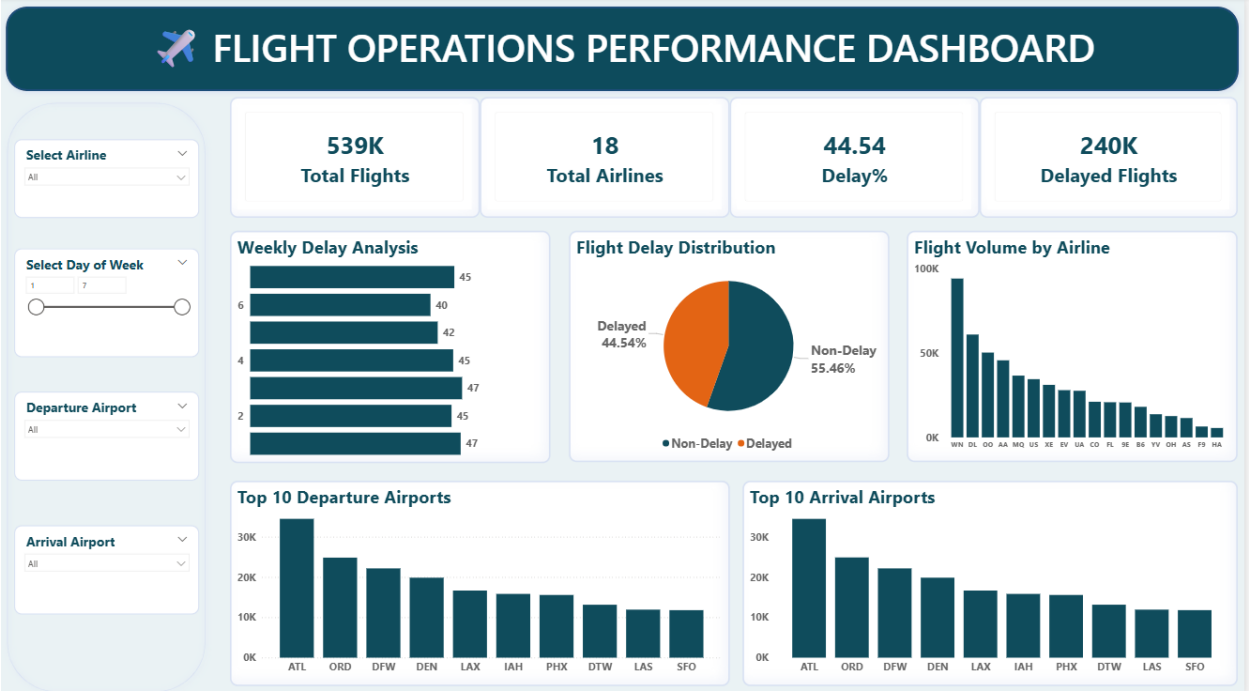
This project successfully analyzed airline operational data using Python, MySQL, and Power BI. Data preprocessing, exploratory data analysis, SQL querying, and dashboard development provided valuable insights into airline performance and flight delays.

The developed interactive dashboard empowers stakeholders to monitor complex flight operations efficiently and supports data-driven decision-making within the aviation industry.

References

1. Airline Flight Operations Public Dataset.
2. Microsoft Power BI Documentation and Design Best Practices (2026).
3. Python Pandas and Matplotlib Documentation.
4. MySQL Reference Manual.

APPENDIX: Flight Operations Performance Dashboard Screenshot



539K

Total Flights

18

Total Airlines

44.54

Delay%

240K

Delayed Flights

Weekly Delay Analysis

45

40

42

45

47

45

47

Flight Delay Distribution

Delayed 44.54%

Non-Delay 55.46%

Flight Volume by Airline

WN

DL

AA

MQ

US

SE

EV

UA

CO

FL

SE

B6

YV

OH

AS

FR

HA

Top 10 Departure Airports

ATL

ORD

DFW

DEN

LAX

IAH

PHX

DTW

LAS

SFO

Top 10 Arrival Airports

ATL

ORD

DFW

DEN

LAX

IAH

PHX

DTW

LAS

SFO